

International Marketing Analytics

Forecasting and time series

WIRTSCHAFTS UNIVERSITÄT WIEN VIENNA UNIVERSITY OF ECONOMICS AND BUSINESS

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Today

Concepts | By hand | How good? | Real data | Python | In class

Theory, Parts 1 to 3

- Trend, season, cycle; stationary or not?
- Simple forecasts and smoothing by hand
- How good is a forecast?

Practice, Parts 4 to 6

- Real data: clothing sales in Australia
- Python: a code ladder in ten steps
- Online quiz and its answers
- Coding exercise and Posit Assistant

Question of the day: how much will we sell next month, and how sure can we be?

Our example: planning demand in two shops

The problem: a marketing manager must plan before demand arrives.

The question: how much will we sell next week, next quarter, next Christmas?

Stock

order before the peak

Staff

shifts for the busy weeks

Media budget

spend when buyers are there

Two tiny shops first: a phone shop (contracts per week) and a chocolate web shop (sales per quarter); real data in Part 4.

Invented round numbers (hand_examples.csv), built for this course

The data: one row per period

Phone shop: new contracts per week

Week	1	2	3	4	5	6	7	8
Contracts	40	44	42	46	50	49	53	54

Chocolate shop: sales per quarter, k EUR

	Q1	Q2	Q3	Q4
Year 1	20	16	20	30
Year 2	24	20	23	35
Year 3	28	24	26	40

period: the time step: week 1 to 8, quarter 1 to 12

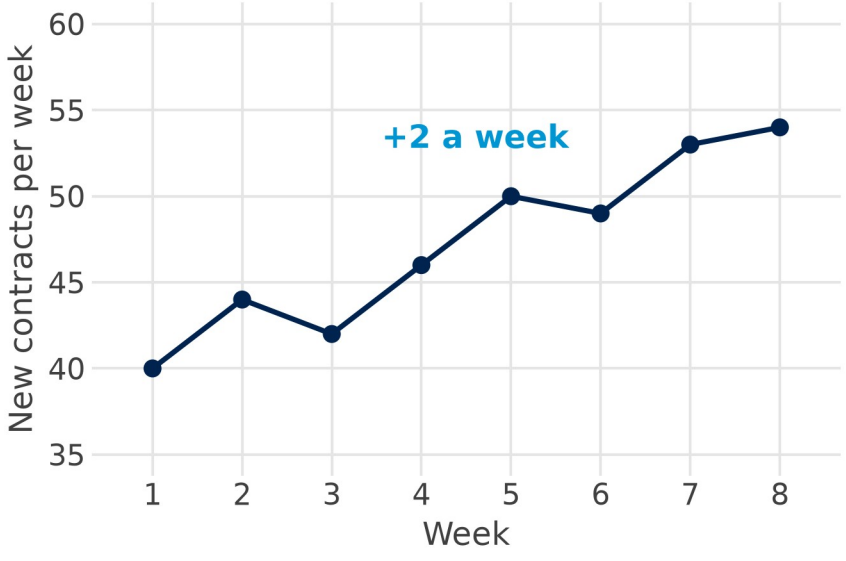
value: what we forecast

rules: time order, equal steps, no gaps

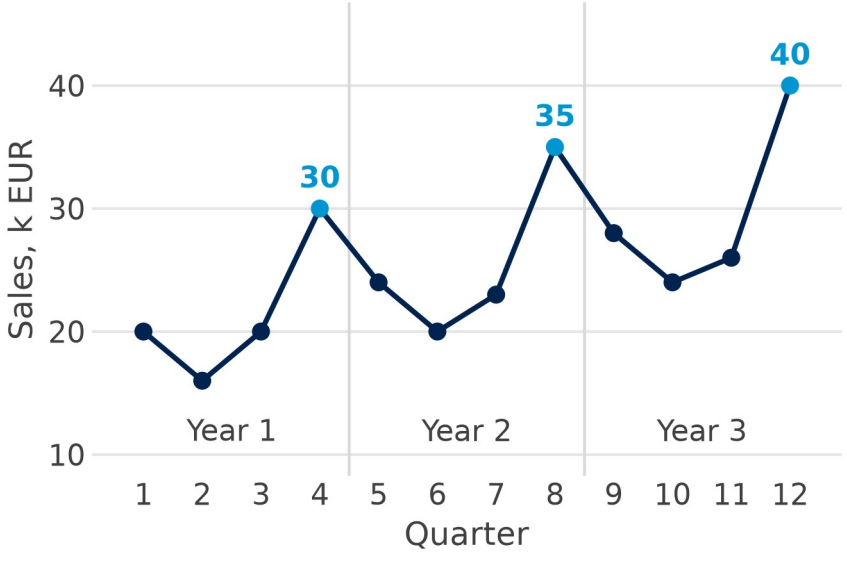
File: `data/forecasting/hand_examples.csv` invented, 20 rows: example, period, value

Invented round numbers (hand_examples.csv)

First look: plot the series



Trend: up about 2 a week



Trend and season: every Q4 peaks

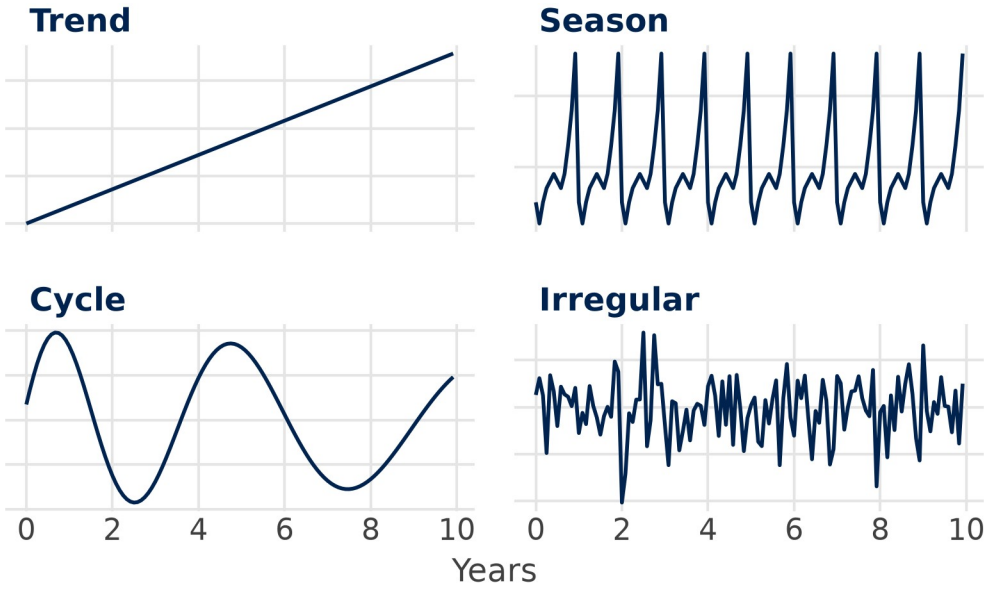
Plot first: the eye finds trend and season before any model.

Invented round numbers (hand_examples.csv)

Time series concepts

Part 1 of 6 · trend, season, cycle, stationarity

Trend, season, cycle and irregular



Trend: the long-run direction

Season: repeats every year, fixed length (Christmas)

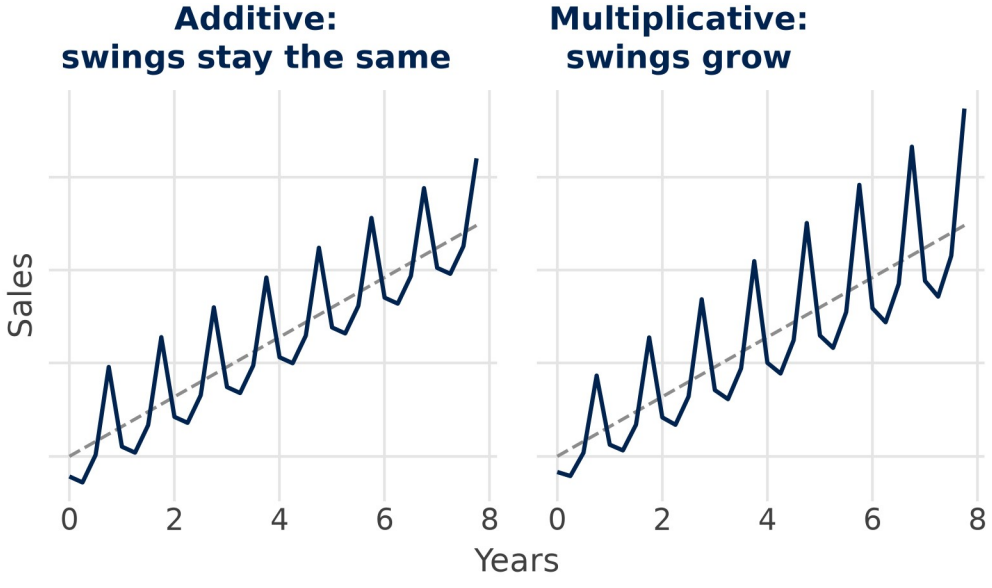
Cycle: ups and downs over years, no fixed length (the economy)

Irregular: what is left: noise, one-off events

Series = trend + season + cycle + irregular (or ×, slide 8): a forecast extends the first three.

Simulated series for illustration

Additive or multiplicative season?



+9.5

chocolate shop: Q4 about 9.5 k EUR above the year mean on average (8.5 to 10.5): additive

× 1.59

clothing (Part 4): December 59 % above an average month, and the swing grows: multiplicative

Multiplicative season: use a multiplicative model or take logs, which makes it additive.

Simulated series for illustration; chocolate shop: invented; clothing: ABS Retail Trade (8501.0), CC BY 4.0

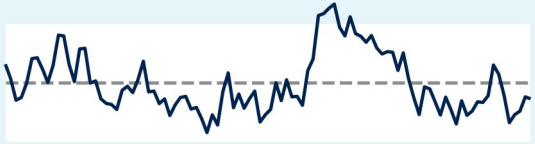
What does stationary mean?

Stationary: the level and the ups and downs stay the same over time; any stretch of the series looks like any other stretch.

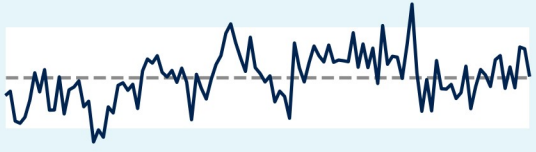
Stationary
noise around a fixed mean



Stationary
wobbles, returns to its mean



Stationary
clothing: change on last year



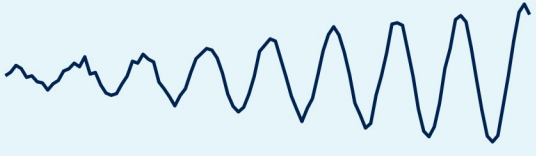
Not stationary
a trend: the level rises



Not stationary
a random walk wanders off



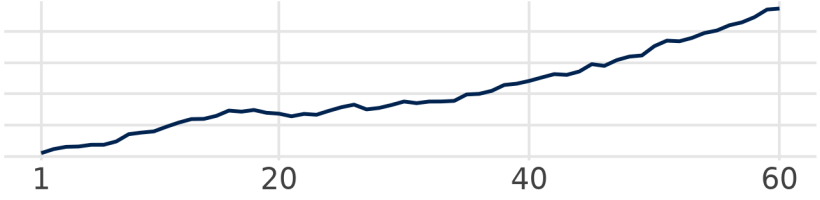
Not stationary
a season whose swings grow



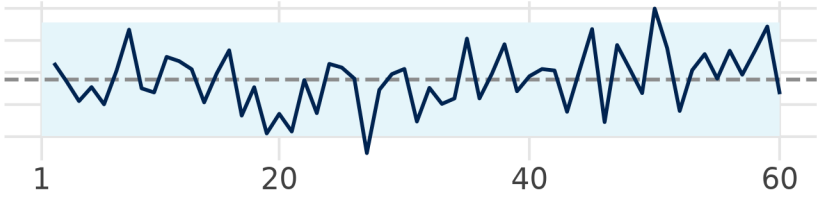
Simulated series for illustration; clothing: ABS Retail Trade (8501.0), CC BY 4.0

Not stationary? Differencing is the fix

A random walk with drift: it rises and wanders



Its changes (this period − the last): stationary



Period

Week	1	2	3	4	5	6	7	8	Average
Contracts	40	44	42	46	50	49	53	54	
Change	–	+4	−2	+4	+4	−1	+4	+1	+2

Differencing: this period − the last; for a season: minus the same period last year. ARIMA-type models need it.

Simulated series for illustration; phone shop: invented round numbers

Simple forecasts by hand

Part 2 of 6 · five methods and smoothing

No change and same change: naive and drift

Week	Contracts	Naive	Error = change
1	40	–	–
2	44	40	+4
3	42	44	−2
4	46	42	+4
5	50	46	+4
6	49	50	−1
7	53	49	+4
8	54	53	+1

54

naive for week 9:
= week 8

56

drift for week 9:
54 + the mean error
(54 − 40) / 7 = +2

Naive's errors are the weekly changes; drift adds their average, +2.

Invented round numbers (hand_examples.csv)

Averages: the mean and a moving average

Week	Contracts	Calculation	Forecast	Error
1	40	–	–	–
2	44	–	–	–
3	42	–	–	–
4	46	(40 + 44 + 42) / 3	42	+4
5	50	(44 + 42 + 46) / 3	44	+6
6	49	(42 + 46 + 50) / 3	46	+3
7	53	(46 + 50 + 49) / 3	48.33	+4.67
8	54	(50 + 49 + 53) / 3	50.67	+3.33

47.25 **mean:** 378 / 8

52 **week 9:** moving average of weeks 6 to 8

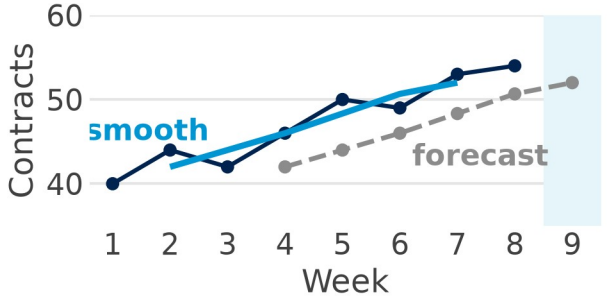
Sum: 40 + 44 + 42 + 46 + 50 + 49 + 53 + 54 = 378

Averages lag behind a trend: every error of the moving average is positive, the forecast is too low.

Invented round numbers (hand_examples.csv)

Rolling averages: smooth or forecast?

Week	Contracts	Centred window	Smooth	Forecast
1	40	–	–	–
2	44	(40 + 44 + 42) / 3	**42**	–
3	42	(44 + 42 + 46) / 3	44	–
4	46	(42 + 46 + 50) / 3	46	**42**
5	50	(46 + 50 + 49) / 3	48.33	44
6	49	(50 + 49 + 53) / 3	50.67	46
7	53	(49 + 53 + 54) / 3	52	48.33
8	54	–	–	50.67
9	?	–	–	52



Smooth: the week before, the week and the week after

Forecast: only the 3 weeks before (slide 13)

Same averages, different place: centre the window to see the trend; look only back to forecast.

Invented round numbers (hand_examples.csv)

Exponential smoothing, step by step

Week	Contracts	Level	Calculation (α 0.5)
1	40	40	start: the first value
2	44	42	0.5 × 44 + 0.5 × 40
3	42	42	0.5 × 42 + 0.5 × 42
4	46	44	0.5 × 46 + 0.5 × 42
5	50	47	0.5 × 50 + 0.5 × 44
6	49	48	0.5 × 49 + 0.5 × 47
7	53	50.5	0.5 × 53 + 0.5 × 48
8	**54**	**52.25**	**0.5 × 54 + 0.5 × 50.5**

Rule

```
new level = α × value
          + (1 − α) × old level
forecast = last level
```

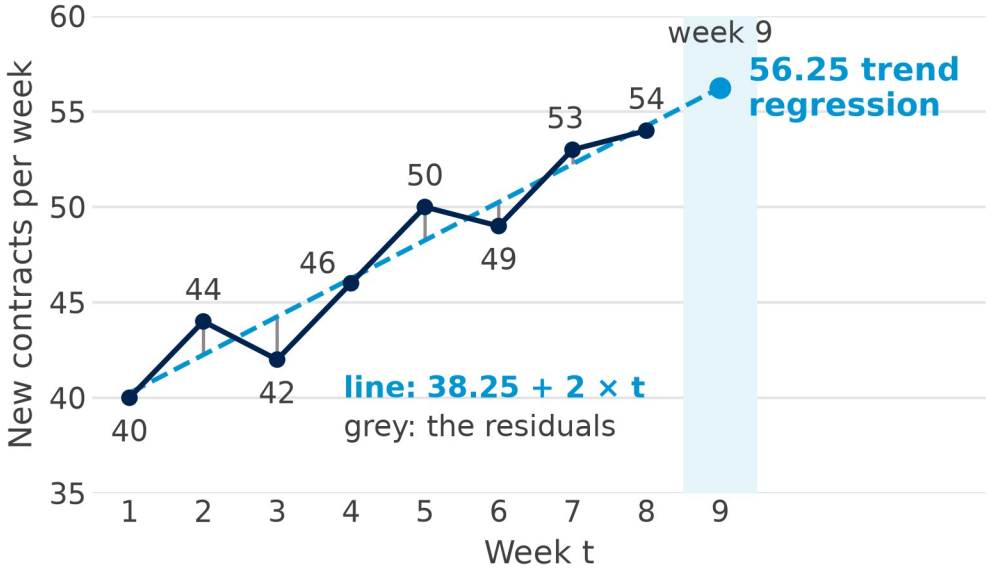
52.25 **forecast** for week 9 (α 0.5)

47.89 **α 0.2:** slower
α 1 = naive (54)

Recent weeks count most: α near 1 follows the data fast, α near 0 averages.

Invented round numbers (hand_examples.csv)

Regression on time: the trend line



By hand

$\bar{t} = 4.5 \quad \bar{y} = 47.25$

$b1 = \Sigma(t - \bar{t})(y - \bar{y}) \,/\, \Sigma(t - \bar{t})^2$

$= 84 \,/\, 42 = 2$

$b0 = \bar{y} - b1 \times \bar{t}$

$= 47.25 - 2 \times 4.5 = 38.25$

week 9: $38.25 + 2 \times 9 = 56.25$

0.926

R^2 = 1 − 13.5 / 181.5
the line fits well

The trend line uses every week: it assumes the trend goes on – and that the errors are independent (slide 32).

Invented round numbers (hand_examples.csv)

The trend line by hand, week by week

Week t	Contracts y	$t - \bar{t}$	$y - \bar{y}$	$(t - \bar{t})(y - \bar{y})$	$(t - \bar{t})^2$	Fitted	Residual
1	40	−3.5	−7.25	+25.375	12.25	40.25	−0.25
2	44	−2.5	−3.25	+8.125	6.25	42.25	+1.75
3	42	−1.5	−5.25	+7.875	2.25	44.25	−2.25
4	46	−0.5	−1.25	+0.625	0.25	46.25	−0.25
5	50	+0.5	+2.75	+1.375	0.25	48.25	+1.75
6	49	+1.5	+1.75	+2.625	2.25	50.25	−1.25
7	53	+2.5	+5.75	+14.375	6.25	52.25	+0.75
8	54	+3.5	+6.75	+23.625	12.25	54.25	−0.25
Sum	**378**	**0**	**0**	**84**	**42**	**378**	**0**

Means: $\bar{t} = 4.5$, $\bar{y} = 47.25$. Fitted = 38.25 + 2 × t; residual = y − fitted.

b1 = 84 / 42 = 2, b0 = 47.25 − 2 × 4.5 = 38.25: two sums give the trend line.

Invented round numbers (hand_examples.csv)

Season by hand: the chocolate shop

Sales, k EUR	Q1	Q2	Q3	Q4	Year mean
Year 1	20	16	20	**30**	21.5
Year 2	24	20	23	**35**	25.5
Year 3	28	24	26	**40**	29.5
Season: average gap	**−1.5**	**−5.5**	**−2.5**	**+9.5**	**sum 0**

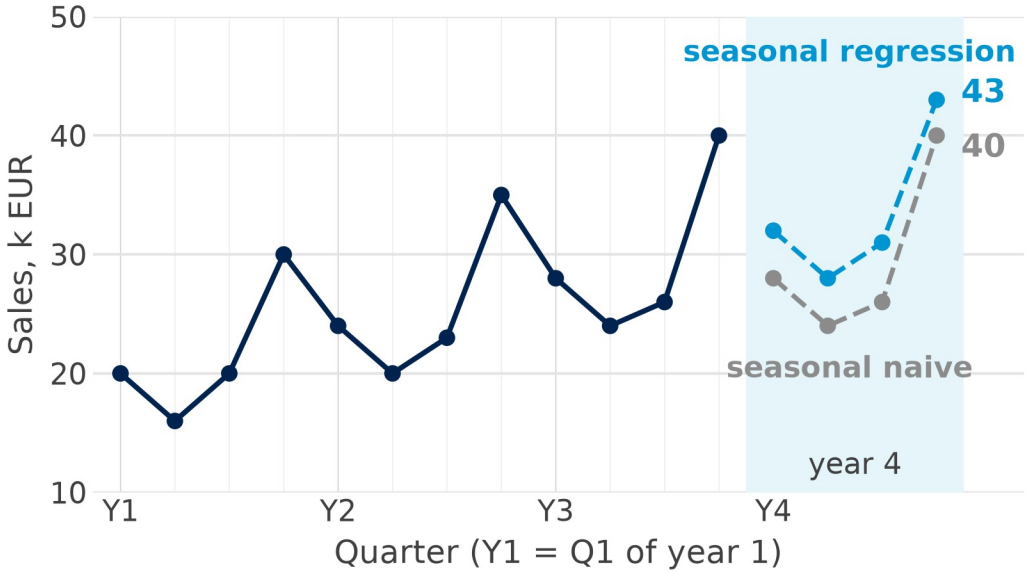
Q4: gaps 30 − 21.5 = +8.5, then +9.5, +10.5
average (8.5 + 9.5 + 10.5) / 3 = +9.5

Trend: year means 21.5 → 25.5 → 29.5
+4 a year, about +1 a quarter (slide 19)

Season = the average gap to the year mean: Q4 sells 9.5 k EUR more than an average quarter.

Invented round numbers (hand_examples.csv)

Two seasonal forecasts for year 4



40 **seasonal naive:**
Q4 of year 4
= Q4 of year 3

43 **seasonal regression:**
$19 + 1 \times 16 + 8$
the trend at t = 16 plus the Q4 effect

Regression, R^2 0.992:
$19 + 1 \times t - 5\ Q2 - 3\ Q3 + 8\ Q4$
Q2, Q3, Q4 = 1 in that quarter, else 0
(Q1 = base)

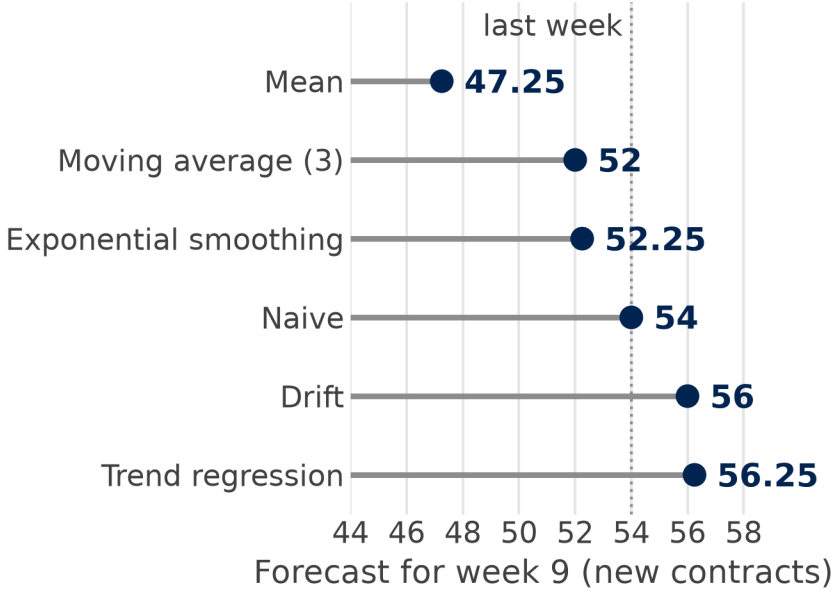
Seasonal naive repeats last year; the regression adds the trend of +1 a quarter.

Invented round numbers (hand_examples.csv)

How good is a forecast?

Part 3 of 6 · hold out, measure, compare

Six forecasts for week 9: which to trust?



1	2	3	4	5	6	7	8
40	44	42	46	50	49	53	54

training period **holdout: weeks 6–8**

Step 1: fit each method on weeks 1–5

Step 2: forecast weeks 6–8, compare with 49, 53, 54

Never at random: test weeks come last.

Six methods, six answers, 9 contracts apart: test each on weeks it has not seen.

Invented round numbers (hand_examples.csv)

Errors and three measures, by hand

Week	Actual	Naive	Error	|Error|	Error²	|Error| / actual
Week 6	49	50	−1	1	1	2.0 %
Week 7	53	50	+3	3	9	5.7 %
Week 8	54	50	+4	4	16	7.4 %
Mean				**2.67**	**8.67**	**5.0 %**

Naive from week 5: 50 for weeks 6, 7 and 8

2.67 **MAE:** mean of |error|

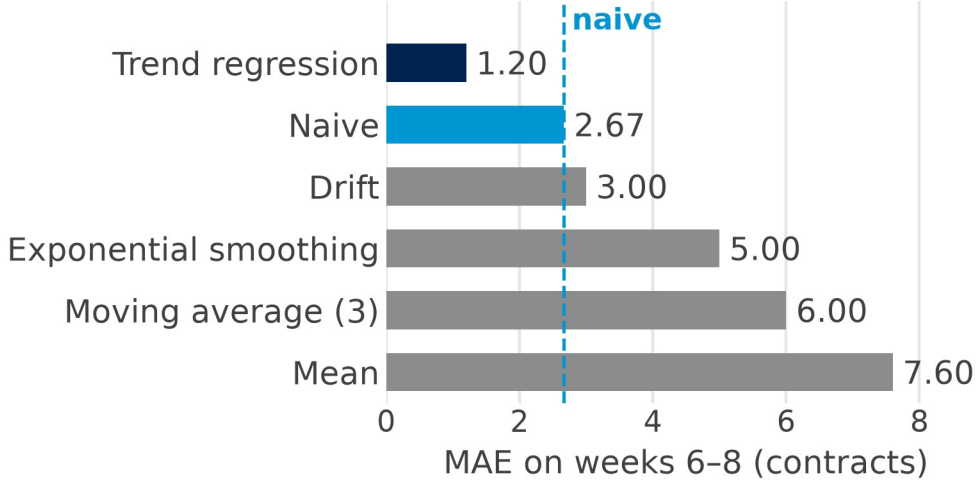
2.94 **RMSE:** √8.67, big errors count more

5.0 % **MAPE:** mean of |error| / actual

Error = actual − forecast: a plus means the forecast was too low.

Invented round numbers (hand_examples.csv)

The ranking: beat the simplest benchmark



1.20 **trend regression:** the only method that beats naive (2.67)

4.00 **chocolate shop, year 3 (MAE, k EUR):** seasonal naive 4.00 vs naive 8.00, so seasonal data need a seasonal benchmark

Forecasts for weeks 6–8, fitted on weeks 1–5: trend regression 37.8 + 2.2 × week = 51, 53.2, 55.4; naive 50; drift 52.5, 55, 57.5; exponential smoothing 47; moving average (3) 46; mean 44.4. RMSE and MAPE give the same order.

Simplest benchmark first: naive, or seasonal naive for seasonal data; keep what beats it.

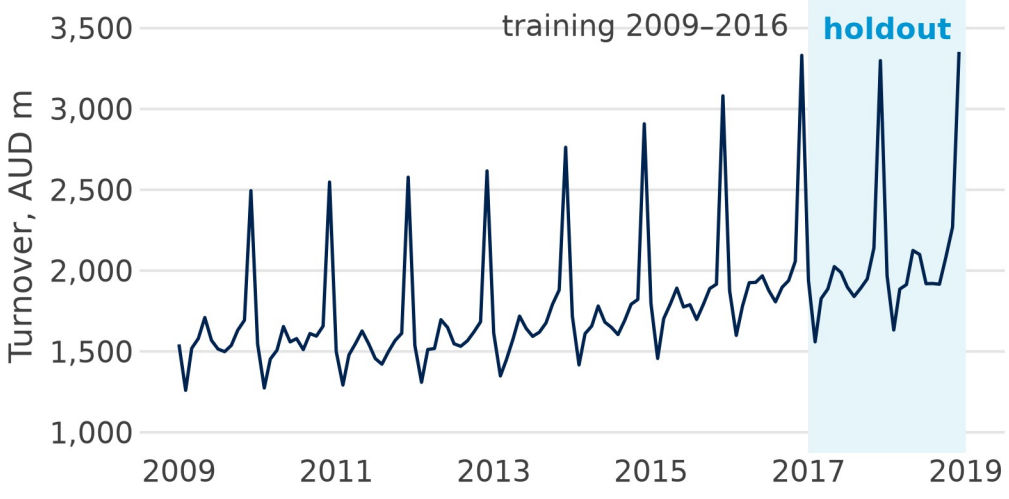
Invented round numbers (hand_examples.csv)

Real data: clothing sales in Australia

Part 4 of 6 · trend, season, holdout, 2019

The data: monthly clothing sales, 2009–2018

The task: a fashion retailer plans stock for 2019 from the market's monthly sales.



Source: Australian Bureau of Statistics (ABS), Retail Trade, open licence (CC BY 4.0)

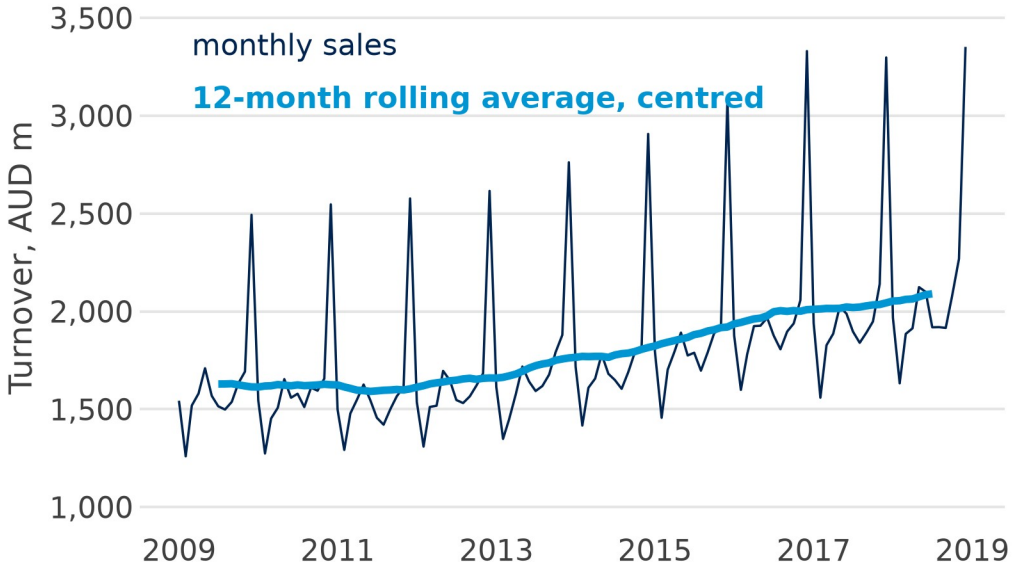
Unit: turnover, AUD million a month, not seasonally adjusted

Period: Jan 2009 to Dec 2018, 120 months: 96 to fit, 24 holdout

File: `data/forecasting/aus_clothing_retail.csv` one row per month: month, value

Source: ABS, Retail Trade, Australia (8501.0), Table 11, clothing, footwear and personal accessories; CC BY 4.0

A 12-month rolling average shows the trend



1,629 — **Jul 2009:** the average month of 2009 (19,546 / 12)

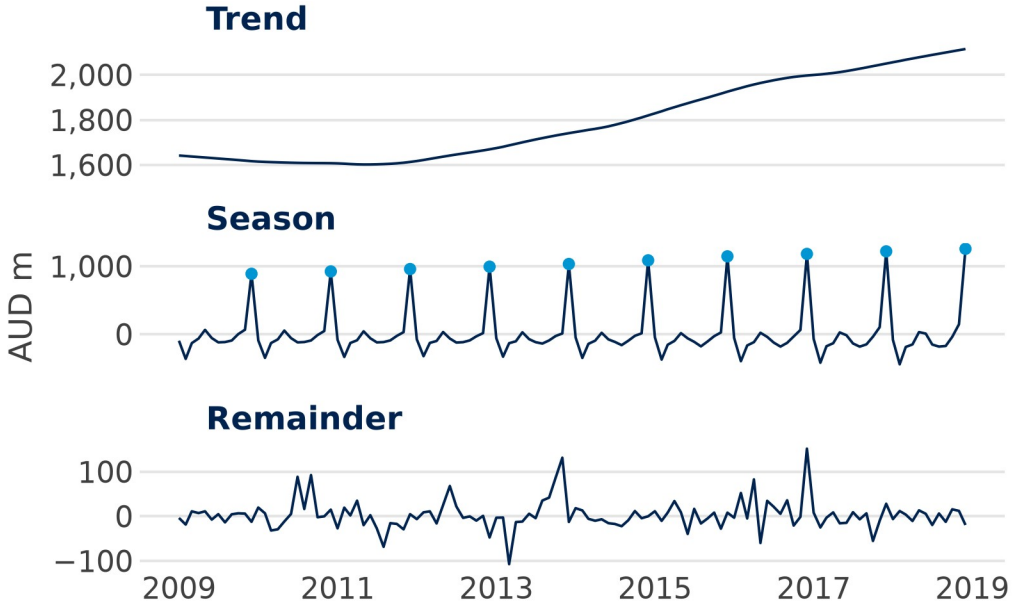
2,090 — **Jul 2018:** the average month of 2018 (25,077 / 12): +28 %

1,591 — **lowest, Jun 2011:** flat to 2012, then growth: the cycle

Twelve months around each month cancel the season: STL (next slide) is a smarter version.

Source: ABS Retail Trade (8501.0), clothing; CC BY 4.0

Trend, season and cycle in the real series



× 1.59 — **December** vs an average month; February × 0.80

+28 % — **2009 to 2018;** flat to 2012 (a cycle), then about +4 % a year

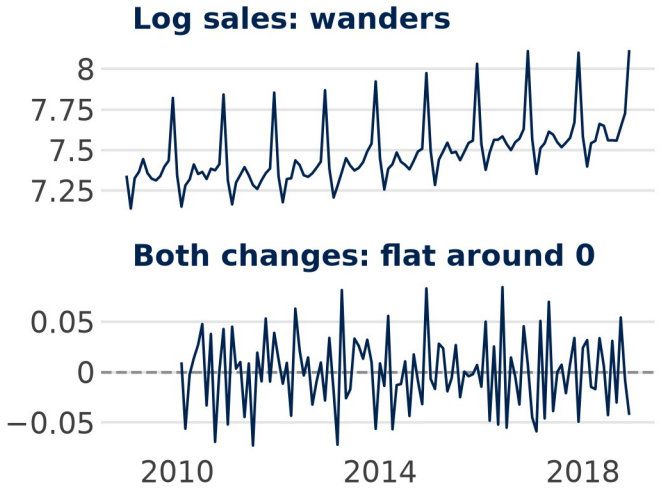
1,257 — **December swing** in 2018, AUD m (2009: 890): it grows with the level

Strong trend and season: a good method must model both.

Source: ABS Retail Trade (8501.0), clothing; CC BY 4.0

Is it stationary? The ADF test

Series	ADF p	Stationary?
Sales (levels)	0.992	no
Log sales	0.987	no
Change on last month	0.011	yes (season stays)
Change on last year	0.003	yes
Both changes, log sales	**< 0.001**	**yes**



ADF in plain words: does it wander without a fixed mean? Small p (below 0.05): no, stationary.

Difference until it is stationary: SARIMA does it for us (next slide).

Source: ABS Retail Trade (8501.0), clothing; CC BY 4.0

Two models that learn trend and season

Holt-Winters / ETS

- Smooths level, trend and season (multiplicative)
- Each has its own weight α, β, γ (slide 15)
- ETS: the same idea as a statistical model, so it also gives bands

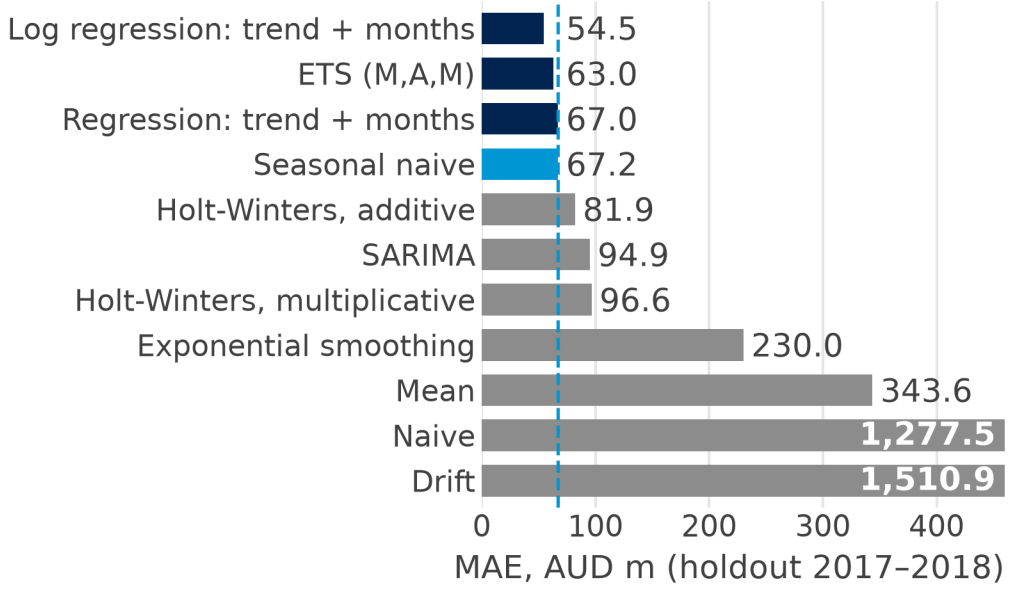
SARIMA

- Differences remove trend and season
- Then it models the memory: how much of last month's and last year's surprise carries on
- Ours: SARIMA$(0,1,1)(0,1,1)_{12}$ on log sales

log sales → change on last month → change on last year → memory of surprises

Both learn trend and season from the data; the holdout decides which forecasts better.

Holdout 2017–2018: beat the seasonal naive



3 of the other 14 methods beat the **seasonal naive** (67.2)

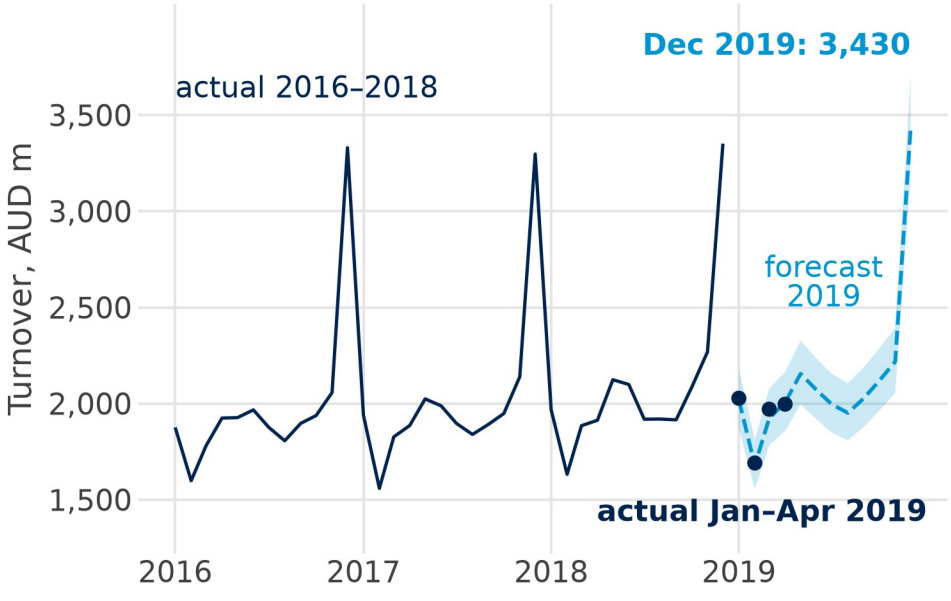
54.5 **best:** log regression, trend + months (MAPE 2.5 %)

3,331 **December trap:** naive repeats Dec 2016 for 24 months (MAE 1,277.5)

On seasonal data the seasonal naive is the benchmark: only three methods beat it.

Source: ABS Retail Trade (8501.0), clothing; CC BY 4.0; holdout 2017–2018

The 2019 forecast with a 95 % band



3,430 — **December 2019;** 95 % band 3,178 to 3,702: should hold the actual in 95 of 100 cases

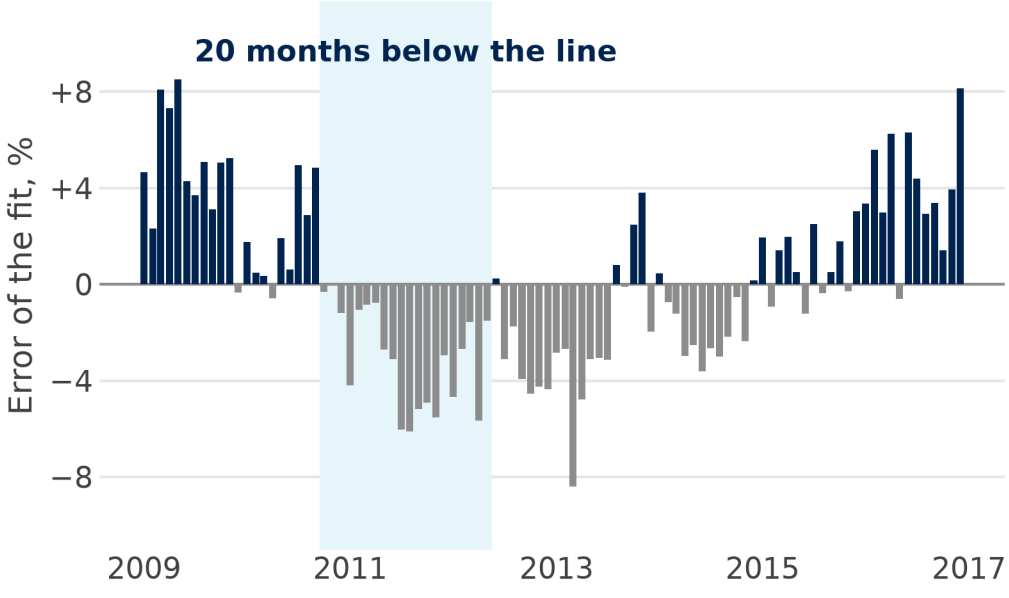
1.0 % — **error Jan–Apr 2019** (MAPE); all 4 inside the band

25,595 — **forecast for 2019** (2018: 25,077)

Refit the winner on all 120 months, then report the forecast with its band.

Source: ABS Retail Trade (8501.0), clothing; CC BY 4.0

Autocorrelation: errors that stick together



0.72 **autocorrelation:** this month's error is like last month's (0 = none)

0.49 **Durbin–Watson:** 2 = no autocorrelation

524 **December band width,** AUD m (regression); SARIMA's widens from 209 in Jan to 589

Errors that stick together: the regression band does not widen with the horizon; for later months use the SARIMA band.

Source: ABS Retail Trade (8501.0), clothing; CC BY 4.0; training months 2009–2016

Forecasting in Python



Part 5 of 6 · a code ladder, one step at a time

The packages and functions

Task	Package	Function
Load and look	polars	`pl.read_csv(); filter(); head()`
Plot the series	plotnine	`ggplot() + geom_line()`
Forecast and MAE	numpy	`np.repeat(); np.resize(); np.mean()`
Smooth	polars	`rolling_mean(12, center=True)`
Exponential smoothing	statsmodels	`tsa.ExponentialSmoothing()`
Regression	statsmodels	`smf.ols("np.log(value) ~ t + C(m)")`
SARIMA	statsmodels	`tsa.SARIMAX()`
Forecast with band	statsmodels	`get_prediction(new).summary_frame()`

Step 1: load and look

forecasting.qmd · Step 1

```
import polars as pl
from plotnine import aes, geom_line, ggplot

# load the monthly sales and keep 2009-2018
file = "../../data/forecasting/aus_clothing_retail.csv"
df = pl.read_csv(file, try_parse_dates=True)
df = df.filter(
    pl.col("month").dt.year().is_between(2009, 2018))
print(df.head())

# plot the series
ggplot(df, aes("month", "value")) + geom_line()
```

df.head()	value
Jan 2009	1,543.2
Feb 2009	1,259.1
Mar 2009	1,518.6

Look first: sales grow, and the December peak grows with them (slide 25).

Step 2: split: fit and holdout

forecasting.qmd · Step 2

```
# the sales as numbers, in time order
y = df["value"].to_numpy()

# fit on the first 96 months, hold out the last 24
train, test = y[:-24], y[-24:]
print(len(train), len(test))
```

Part	Months	Period
train	96	2009–2016
test	24	2017–2018



Test on the future: fit on 2009–2016, forecast 2017–2018; never split at random.

Step 3: naive and seasonal naive

forecasting.qmd · Step 3

```
import numpy as np

# naive: repeat the last training month 24 times
naive = np.repeat(train[-1], 24)

# seasonal naive: repeat the last training year twice
snaive = np.resize(train[-12:], 24)
print(naive[:12])
print(snaive[:12])
```

2017	naive	snaive
Jan	3,330.5	1,875.7
Jun	3,330.5	1,967.1
Dec	3,330.5	3,330.5

Naive forecasts a Christmas every month; the seasonal naive repeats 2016.

Step 4: how good? The MAE

forecasting.qmd · Step 4

```
# MAE: the mean absolute error on the holdout
mae_snaive = np.mean(np.abs(test - snaive))
mae_naive = np.mean(np.abs(test - naive))
print(mae_snaive, mae_naive)
```

Method	MAE
snaive	67.2
naive	1,277.5



The December trap: naive misses by 1,277.5 a month; beat the seasonal naive.

Step 5: smooth: a rolling average

forecasting.qmd · Step 5

```
# 12-month rolling average, centred: the trend
smooth = df.with_columns(
    trend=pl.col("value").rolling_mean(12, center=True))
print(smooth.drop_nulls())

# the sales and the trend in one chart
chart = ggplot(smooth, aes("month", "value")) + geom_line()
chart + geom_line(aes(y="trend"), color="blue")
```

trend	AUD m
Jul 2009	1,628.8
Jul 2018	2,089.8



The season cancels out: the trend is flat to 2012, then grows (+28 %).

Step 6: exponential smoothing

forecasting.qmd · Step 6

```
import statsmodels.tsa.api as tsa

# Holt-Winters: a trend and a season that grows
hw = tsa.ExponentialSmoothing(
    train, trend="add", seasonal="mul",
    seasonal_periods=12,
    initialization_method="estimated").fit()
hw_fc = hw.forecast(24)
mae_hw = np.mean(np.abs(test - hw_fc))
print(mae_hw)
```

Method	MAE
Holt-Winters	96.6
snaive	67.2

Smarter is not always better: Holt-Winters loses to the seasonal naive.

Step 7: regression: trend and months

forecasting.qmd · Step 7

```
import statsmodels.formula.api as smf

# t counts the months 1-120, m is the month 1-12
d = df.with_columns(
    t=pl.int_range(1, pl.len() + 1),
    m=pl.col("month").dt.month()).to_pandas()
# log sales on trend + month dummies, first 96 months
reg = smf.ols("np.log(value) ~ t + C(m)",
              data=d[:-24]).fit()
reg_fc = np.exp(reg.predict(d[-24:]))
mae_reg = np.mean(np.abs(test - reg_fc))
print(mae_reg)
```

Method	MAE
regression	54.5
snaive	67.2

The first winner: trend + months on log sales beats the seasonal naive.

Step 8: SARIMA

forecasting.qmd · Step 8

```
# SARIMA on log sales: a change on last month
# and on last year, and the memory of surprises
sar = tsa.SARIMAX(
    np.log(train), order=(0, 1, 1),
    seasonal_order=(0, 1, 1, 12)).fit(disp=0)
sar_fc = np.exp(sar.forecast(24))
mae_sar = np.mean(np.abs(test - sar_fc))
print(mae_sar)
```

Method	MAE
SARIMA	94.9
snaive	67.2

Test, do not assume: SARIMA also loses to the seasonal naive here.

Step 9: compare the four

forecasting.qmd · Step 9

```
# the four MAEs side by side
pl.DataFrame({
    "method": ["snaive", "hw", "reg", "sarima"],
    "MAE": [mae_snaive, mae_hw, mae_reg, mae_sar]})
```

method	MAE
snaive	67.2
hw	96.6
reg	54.5
sarima	94.9

The regression wins: the same ranking as slide 30.

Step 10: forecast 2019

forecasting.qmd · Step 10

```
# refit the winner on all 120 months
reg_all = smf.ols("np.log(value) ~ t + C(m)",
                  data=d).fit()

# 2019: t = 121 to 132, m = 1 to 12
new = pl.DataFrame({"t": range(121, 133),
                    "m": range(1, 13)}).to_pandas()

# the forecast and its 95 % band, back from logs
band = reg_all.get_prediction(new).summary_frame()
np.exp(band[["mean", "obs_ci_lower", "obs_ci_upper"]])
```

2019	mean	low	high
Jan	2,020.4	1,871.9	2,180.6
Dec	3,430.2	3,178.2	3,702.2

Plan December with 3,430 AUD m: the 95 % band runs from 3,178 to 3,702.

In-class exercise

Part 6 of 6 · quiz, exercise, Posit Assistant

Online quiz: forecasting in five questions

1 Sales: 50, 54, 58. The drift (same change) forecast for the next period is …
A 58 B 62 C 54

2 Exponential smoothing with α 0.5: level 40, new value 50. The new level is …
A 45 B 50 C 42

3 Holdout errors: −2, +2, +4. The MAE is …
A 1.33 B 4.00 C 2.67

4 Monthly sales with a December peak. The benchmark to beat is …
A naive (last month) B seasonal naive (same month last year) C the mean

5 ADF test on sales levels: p 0.99. The series is …
A stationary B white noise C not stationary: difference it

3 minutes alone, then compare with your neighbour.

Answers: in class

Your turn: forecast café sales in Python

Pairs: 35 minutes	Plenary: 5 minutes	Posit Assistant: 10 minutes

1. **By hand:** café bookings: five forecasts for week 7, MAE on weeks 5–6 (Steps 3, 4; Extra A)
2. **Look:** plot 2009–2018; trend, season, ADF (Step 1; Extra C)
3. **Test:** five methods on the 2017–2018 holdout: MAE, RMSE, MAPE (Steps 3, 6–9)
4. **Forecast:** 2019 with the winner and SARIMA, 95 % bands; check Jan–Apr (Step 10; Extra D)

Deliverable: one slide per pair: the 2019 forecast and two sentences.

File: `sessions/01-foundations/forecasting_exercise.qmd` (reuse: forecasting.qmd)

Then: the same tasks with Posit Assistant

How: code first, then open Posit Assistant, type @forecasting_exercise.qmd, one task per prompt

Prompt for task 1

```
@forecasting_exercise.qmd Task 1. Café
bookings 20, 24, 22, 26, 27, 31: naive,
drift, mean, moving average (3) and SES
(α 0.5) for week 7, step by step.
```

Prompt for task 2

```
Task 2. Plot aus_cafes_retail.csv for
2009-2018 with plotnine. Describe trend
and season; run the ADF test on levels
and on the first difference.
```

Prompt for task 3

```
Task 3. Hold out 2017-2018. Compare
seasonal naive, Holt-Winters, SARIMA and
two regressions by MAE, RMSE and MAPE
in a great_tables table.
```

Prompt for task 4

```
Task 4. Forecast 2019 with the best
method and with SARIMA, 95 % bands.
Which band would you trust for stock
planning, and why?
```

Check every number against your own code: you own the result.

Open data for forecasting

ABS Retail Trade

Monthly turnover in Australia since 1982, by state and industry

Licence: CC BY 4.0

Eurostat

Monthly retail trade and tourism nights by EU country

Licence: free reuse, cite the source

M5 (Walmart)

Daily unit sales of 3,049 products in 10 US stores, 2011–2016

Licence: competition data: check the terms

The licence decides: commit only data you may share; link the rest.

Links and loading code: `forecasting.qmd` (appendix)

Sources: ABS, Eurostat and M5 dataset pages (links in forecasting.qmd)

Key takeaways

1. **Plot and smooth first:** trend, season, cycle; additive or multiplicative; stationary or not?
2. **Start simple:** naive, drift, averages, smoothing and a trend line: all on a calculator
3. **Test on the future:** hold out the last periods; compare MAE, RMSE and MAPE
4. **Beat the simplest benchmark:** on seasonal data, the seasonal naive
5. **Report the band, check the errors:** autocorrelated errors make a band too narrow

Next: run the ten steps of sessions/01-foundations/forecasting.qmd, then the exercise.

References

Forecasting methods

Hyndman, R. J. & Athanasopoulos, G. (2021). Forecasting: Principles and Practice (3rd ed.). OTexts. otexts.com/fpp3

Box, G. E. P., Jenkins, G. M., Reinsel, G. C. & Ljung, G. M. (2015). Time Series Analysis: Forecasting and Control (5th ed.). Wiley.

Hyndman, R. J. & Koehler, A. B. (2006). Another look at measures of forecast accuracy. International Journal of Forecasting, 22(4), 679–688.

Cleveland, R. B. et al. (1990). STL: A seasonal-trend decomposition procedure based on loess. Journal of Official Statistics, 6(1), 3–73.

Competitions, data and software

Makridakis, S., Spiliotis, E. & Assimakopoulos, V. (2020). The M4 Competition: 100,000 time series and 61 forecasting methods. International Journal of Forecasting, 36(1), 54–74.

Makridakis, S. et al. (2022). M5 accuracy competition: Results, findings, and conclusions. International Journal of Forecasting, 38(4), 1346–1364.

Australian Bureau of Statistics (2019). Retail Trade, Australia, April 2019, cat. no. 8501.0, Table 11. CC BY 4.0.

Seabold, S. & Perktold, J. (2010). statsmodels: Econometric and statistical modeling with Python. Proceedings of the 9th Python in Science Conference.

Free textbook: `otexts.com/fpp3` · Python edition: `otexts.com/fpppy`

Thank you. Questions and discussion

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